Team Number: 012-6
Team Name: AudioFilers
Team Members: Matthew Aldridge, Dylan Long, Greg Lowman, Ian Martin, Devang Pandey, Vera Zaric
Application Name: Daily Dev Mix
Application Description: This application aims to record metadata from a Spotify session and provide suggested personalized playlists. The user starts a session when they are ready to listen to music and ends it when they want the recording to stop. They also select a session type to define what they are doing (e.g., studying, road trip, workout, etc.). Using the received metadata and Spotify APIS, we should have ample data to work with.
The output should include truly personalized playlists. Currently, there are already playlists based on activities on Spotify, but they are overly generalized and do not match the individual listening experience. Our API aims to create personalized playlists based on the music listened to during the session, so users can enjoy music they actually want to hear while engaging in their activity.

Audience: This is for spotify users who don't align with the listening experience of the generalized activity playlists Spotify generates.

Vision Plan: For Spotify users who want more personalized playlists. The Daily Dev Mix is an API that works to build playlists for the user that are personalized around activities (ie. road trips, studying, or exercising). Unlike the native playlist suggestions, our product will be much less general and tuned to the individual.

Communication Plan:
We will be using a Discord server for group communication and coordination.

Team meets: Right after recitation (Thursdays at 2:30)
Modality: In person / On Zoom if unavailable

TA meets: Thursdays 2:15-2:30
Modality: In-person after Lab

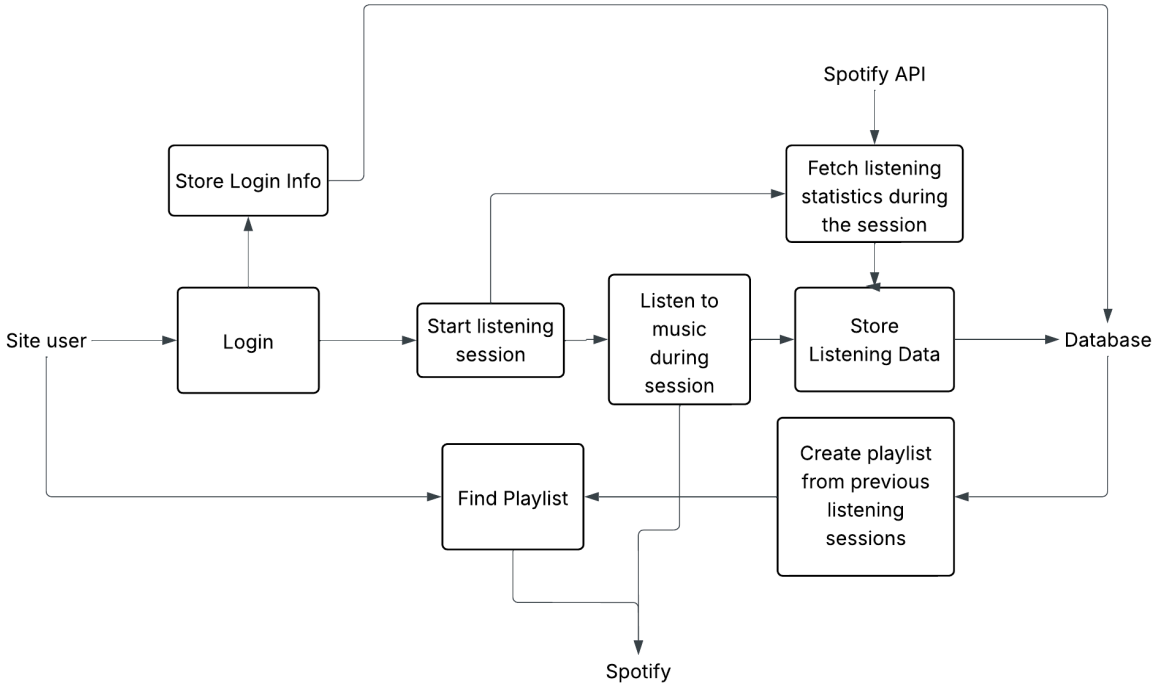
Wire Frame and Use Case Diagram: Included at the end of the document

Extra Credit:

1. Spotify API Rate Limiting and Scope Changes
    a. Severity: High
    b. We may hit rate limits, causing the app to crash or lag for users. Additionally, Spotify could restrict access to specific metadata fields at any time.
    c. Mitigation:
        i. Optimize Calls: Make it so that our API requests and cache data locally so we aren't pinging Spotify for the same information repeatedly.

        ii. Exponential Backoff: Implement an algorithm that automatically slows down requests if the API returns a "429 Too Many Requests" error.
2. "Cold Start" and Data Sparsity
    a. Severity: Medium
    b. If the first few personalized playlists are inaccurate because the user has only recorded 20 minutes of an activity, they may abandon the app before the algorithm matures.
    c. Mitigation:
        i. Hybrid Recommendation: Use a warm-up period where you supplement unique session data with the user's general top tracks or recently played from Spotify until you have enough specific session data to take over.
        ii. Seed Playlists: Ask users to select 3-5 anchor songs when they create a new session type for the first time.
3. User Forgetfulness (Session Fatigue)
    a. Severity: Medium
    b. The biggest risk to data integrity is the human element. Users may start a study session, finish their work, and go to the gym without ending the session. This pollutes your metadata, mixing Lo-Fi beats with heavy metal under the study tag.
    c. Mitigation:
        i. Smart Reminders: Send a push notification if a session has been active for an unusually long time.
        ii. Auto-Stop Logic: Implement a feature that ends a session automatically after 30 minutes of silence or inactivity from the Spotify player.
4. Data Privacy and Security
    a. Severity: High
    b. You are collecting specific behavioral data (where people are, when they work out, what their daily routines look like). A data breach could expose sensitive habits of your users, and you must stay compliant with regulations like GDPR or CCPA.
    c. Mitigation:
        i. Minimalism: Only store the metadata absolutely necessary for the algorithm. Avoid storing sensitive PII (Personally Identifiable Information) on your own servers.
        ii. Encryption: Use OAuth 2.0 for all Spotify authentications so you never actually see or store user passwords.
5. Algorithmic Echo Chambers
    a. Severity: Low/Medium
    b. If your app only recommends music based on what a user has already listened to during a session, the playlists might become repetitive. The user might get bored because the personalized workout playlist sounds exactly like their last 5 workout history.
    c. Mitigation:

i. The "Exploration" Variable: Introduce a discovery slider that allows users to decide how much new music vs. familiar music they want in their generated playlist.
ii. Collaborative Filtering: Use the session data of other users with similar tastes to suggest songs the primary user hasn't heard yet but fits the vibe.



Wireframe
Link to Figma Design: https://woven-trial-07775303.figma.site/

Daily Dev Mix

Track your listening sessions and get truly personalized playlists based on your actual music preferences

Choose your vibe

- Studying
- Workout
- Road Trip
- Relaxing
- Party
- Deep Focus

Start Listening Session

Quick access

Your Playlists

Listening History

Created with Figma Make Remix

LIVE SESSION

Studying

End Session

1	0m 6s	73%	51%
Tracks	Duration	Energy	Vibe

Now Playing

Blinding Lights
The Weeknd
After Hours

Energy	Dance	Mood	Tempo	Acoustic
73%	51%	33%	171	0%

Session History

1	Blinding Lights	The Weeknd	3:20

Created with Figma Make Remix

Your Library

Personalized playlists generated from your sessions

Study Zone Mix

Your personalized study playlist based on focus sessions

4 songs • 12 min

Workout Energy

High-energy tracks from your best workout sessions

4 songs • 10 min

AI-Powered Personalization

Each playlist is generated using advanced audio analysis and machine learning. We analyze energy levels, danceability, mood (valence), tempo, and acoustic features from the tracks you listened to during your session. Our algorithm then finds similar tracks that match your unique taste - creating truly personalized mixes, not just generic activity-based recommendations.

Created with Figma Make Remix

LIVE SESSION

Party

End Session

1	0m 9s	73%	51%
Tracks	Duration	Energy	Vibe

Now Playing

Blinding Lights

The Weeknd

After Hours

Energy	Dance	Mood	Tempo	Acoustic
73%	51%	33%	171	0%

Session History

1 **Blinding Lights** The Weeknd 3:20